

ALZ 3822

SUBMERSIBLE PRESSURE TRANSMITTER WITH DETACHABLE PROBE



DESCRIPTION

ALZ 3822 is a general purpose submersible hydrostatic level transmitter with detachable probe, allowing quick and easy removal for service or replacement. High-quality silicon piezoresistive sensor, plastic housing and a hydrometric cable with the ventilation tube make ALZ 3822 a reliable solution for a wide variety of industrial level measurement applications.

SPECIFICATIONS

Ranges: 1 mH₂O up to 100 mH₂O

Basic accuracy: $\pm 0.25\%$

Probe: Ø35 mm, plastic, detachable from the cable for easy service

Outputs: 4...20 mA; 0...20 mA; 0...10 V; 0...5 V; RS-485 / Modbus RTU; HART®

Sensor: silicon piezoresistive

Media temperature: -20...+50 °C

HART® is a registered trademark of the HART Communication Foundation.

APPLICATIONS

Water, wastewater

Fuels, oils

Tanks, wells

Pump protection

TECHNICAL SPECIFICATIONS

MEASURING RANGES

Pressure range, bar	Level, mH ₂ O	Overpressure, bar	Pressure range, bar	Level, mH ₂ O	Overpressure, bar
0...0.1	0...1	1	0...1.6	0...16	6
0...0.16	0...1.6	1	0...2.5	0...25	6
0...0.25	0...2.5	1	0...4	0...40	15
0...0.4	0...4	1	0...6	0...60	15
0...0.6	0...6	3	0...10	0...100	30
0...1	0...10	3	-	-	-

PERFORMANCE	P > 0.4 bar	P ≤ 0.4 bar
Accuracy, % of span*	≤ ±0.25 (standard) / 0.2 (optional)	≤ ±0.5 (standard)
Temperature effect (% of span / 10 °C)	≤ ±0.1	≤ ±0.2
Compensated range	-20...+50 °C	0...+50 °C
Power supply effect	≤ ±0.05% of span / 10 V	
Load resistance effect	≤ ±0.05% of span / kOhm (transmitters with current output)	
Long-term stability	≤ ±0.1% of span / year	
Response time (10...90%)	≤ 1 ms	

* Accuracy includes non-linearity, hysteresis and non-repeatability.

OPERATING CONDITIONS

Medium temperature (depends on seal)	-20...+50 °C
Cable service temperature (static applications)	PVC: -20...+70 °C PUR: -50...+70 °C
Storage temperature	-20...+50 °C
Vibration resistance	10 g RMS, 25–2000 Hz
Shock resistance	100 g / 11 ms
Service life	> 100 x 10 ⁶ cycles

MECHANICAL SPECIFICATIONS

Housing	PVC
Seal	FKM; EPDM; NBR
Diaphragm	stainless steel 316L (1.4435)
Cable external jacket	PVC, PUR, FEP
Ingress protection	IP69K

ELECTRICAL SPECIFICATIONS

Output signal	Power supply, U _s	Load resistance, R	Power consumption
4...20 mA / 2-wire	12...36 V	$\leq [(U_s - 12 \text{ V}) / 0.02 \text{ A}] \text{ Ohm}^*$	$\leq 26 \text{ mA}$
4...20 mA / HART®			
4...20 mA / 3-wire		$\leq 500 \text{ Ohm}$	
0...20 mA / 3-wire		$\geq 10 \text{ kOhm}$	$< 7 \text{ mA}$
0...10 V / 3-wire			
0...5 V / 3-wire	5 V	$\geq 5 \text{ kOhm}$	$\leq 2 \text{ mA}$
0.5...4.5 V / 3-wire	6...15 V		$\leq 7 \text{ mA}$
0.5...4.5 V / 3-wire	12...36 V	-	
RS-485 / Modbus RTU			

* For output signal 4...20 mA / HART® minimum load resistance for digital communication: 250 Ohm.

ELECTRICAL CONNECTIONS / PIN ASSIGNMENT

Circuits		Cable gland wires
2-wire	power +	white (red)
	power -	brown (blue)
	ground	yellow-green
3-wire	power +	white (red)
	power -	brown (blue)
	signal +	green (black)
	ground	yellow-green
RS-485	power +	white (red)
	power -	brown (blue)
	A	yellow
	B	green (black)
	ground	yellow-green
HART®	power +	white
	power -	brown
	ground	yellow-green

DIMENSIONS (mm)



ORDERING CODE

ALZ 3822		-X	-XXXX	-X	-X	-X	-XX	// X	-XXXX
UNIT OF MEASUREMENT									
	bar	B							
	kg/cm ²	S							
	mH ₂ O	W							
	kPa	H							
Other (specify when ordering)		X							
UPPER RANGE LIMIT (URL)									
bar, kg/cm ²		mH ₂ O		kPa					
0.1	0100	1	1000	10	1001				
0.16	0160	1.6	1600	16	1601				
0.25	0250	2.5	2500	25	2501				
0.4	0400	4	4000	40	4001				
0.6	0600	6	6000	60	6001				
1	1000	10	1001	100	1002				
1.6	1600	16	1601	160	1602				
2.5	2500	25	2501	250	2502				
4	4000	40	4001	400	4002				
6	6000	60	6001	600	6002				
10	1001	100	1002	1000	1003				
Other	XXXX	Other	XXXX	Other	XXXX				
ACCURACY									
0.25% (P > 0.4 bar) (standard)				C					
0.5% (P ≤ 0.4 bar) (standard)				D					
0.2% (P > 0.4 bar)				B					
Other (specify when ordering)				X					
OUTPUT SIGNAL									
4...20 mA / 2-wire				A					
4...20 mA / 3-wire				B					
0...20 mA / 3-wire				C					
0...10 V / 3-wire				D					
0...5 V / 3-wire				E					
0.5...4.5 V / 3-wire, U _s = 5 V				R					
0.5...4.5 V / 3-wire, U _s = 6...15 V				K					
RS-485 / Modbus RTU				M					
4...20 mA / HART®				H					
Other (specify when ordering)				X					
SEALS									
FKM				F					
NBR				N					
EPDM				E					
Other (specify when ordering)				X					
VERSION									
Standard				00					
Other (specify when ordering)				XX					

ORDERING CODE (CONTINUED)

ALZ 3822	-X	-XXXX	-X	-X	-X	-XX	// X	-XXXX
CABLE EXTERNAL JACKET								
							PVC	P
							PUR	U
							FEP	T
							Other (specify when ordering)	X
CABLE LENGTH								
Any length in meters (e.g., 010M means 10 meters)								XXXM

Example: ALZ 3822-W-6000-D-A-F-00 // P-010M

ACCESSORIES

				
Anchoring clamp for submersible transmitter	PZ 1024 Power supply unit	BZ 05 / BZ 10 Dry air junction box for submersible transmitters		